CASE 1 - ELECTRICITY PRICE FORECASTING

BACKGROUND

As the global transition to renewable energy gathers pace, individual power generation is becoming an increasingly viable option. Many homeowners and small-scale entrepreneurs are installing solar panels, wind turbines, and battery storage systems. This decentralized energy generation model has the potential to reshape energy markets, not only from an infrastructural standpoint but also from a financial trading perspective, especially for investment banks that are always on the lookout for emerging market trends. Individuals feeding excess power back into the grid present both a challenge and an opportunity. For investment banks, accurately forecasting these feed-in times and quantities is paramount. It aids them in creating financial products, derivatives, and trading strategies tailored to this new energy landscape. The crucial question, therefore, is: When is the optimal time for these individual power providers to reintegrate energy for maximum financial return, both for themselves and for the broader financial market?

Emerging technologies are now empowering even lay consumers with insights into the determinants of optimal energy feed-in timings. Such a transparent approach can significantly benefit both individual power producers and the larger grid in terms of efficiency. However, for investment banks, this clarity is doubly beneficial. It allows them to manage risks, devise new financial instruments, and offer guidance to their clientele about potential investment opportunities in this domain. Still, the industry's intricate terminology and multiple influencing factors can convolute this understanding.

The challenge becomes multi-dimensional: Firstly, can we leverage diverse technological tools, like data models and satellite imagery, to demystify real-time and forecasted grid needs? Secondly, can this information be relayed in a straightforward manner to individual power generators, ensuring they make well-informed decisions? Lastly, how can investment banks harness this data to innovate their trading strategies and financial offerings?

In the context of a rapidly evolving energy and financial landscape, how can we devise a transparent forecasting model that not only guides individual power providers on optimal feed-in times but also aids investment banks in navigating the complexities and opportunities of decentralized energy trading?

GOAL

Design a predictive algorithm that determines optimal energy feed-in timings for individual power producers, enabling investment banks to better assess the financial implications and market dynamics of increased decentralized energy contributions to the grid.

SUGGESTED APPROACH

- Technology exploration
    - Explore Transformer-based Timeseries predictions
    - Find available datasources within Norwegian market
        - Historical weather from met.no
- Project Specification
    - Define the requirements of the system.
- System Architecture
    - Describe an overview of how a system can be built.
- Development of Minimum Viable Concept

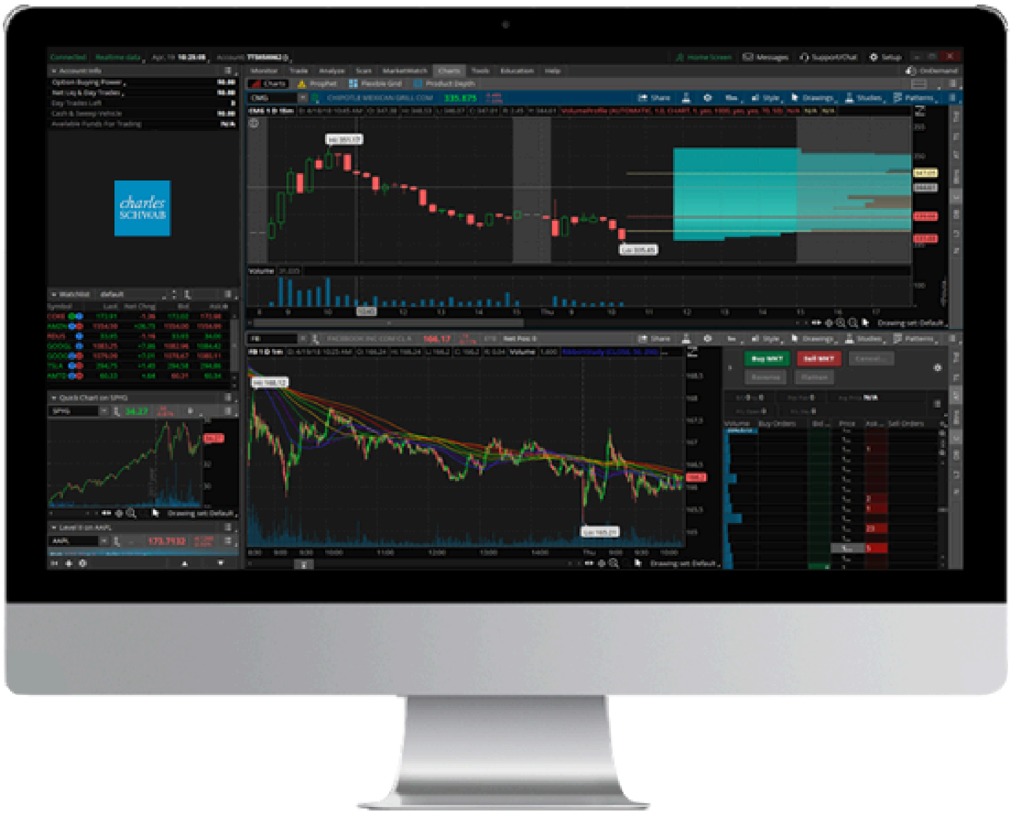
Overview of key data and tools

- https://wegaw.com/swiss-industry-and-science-consortium-on-track-to-optimise-hydropower-production-using-satellite-data/
- https://business.esa.int/projects/defrost-for-hydropower
- https://www.nature.com/articles/s41597-023-02008-2
- https://huggingface.co/docs/transformers/model*doc/time*series_transformer

Open datasets

- https://www.kaggle.com/datasets?search=electricity+price
- https://www.kaggle.com/code/dimitriosroussis/electricity-price-forecasting-with-dnns-eda

CASE 2 - LIVE TRADING PLATFORM



BACKGROUND

The basis for this case is to explore different user cases of The Alpaca platform which is a modern, cloud-based trading platform that provides easy access to market data, order management, and trading APIs. The platform is designed to be highly scalable and customizable, making it an ideal foundation for a wide range of trading applications

PROBLEM DESCRIPTION

Participants in the hackathon will be tasked with creating new solutions based on the Alpaca platform that can help traders make better-informed decisions and execute trades more efficiently in addition to make powerful trading tools more available for a wide set of users. Solutions may include new trading algorithms, automated trading bots, mobile trading applications, or integrations with other financial platforms and services.
One of the key features of the Alpaca platform is its ease of use, and participants in the hackathon will be encouraged to build solutions that are equally user-friendly. Additionally, solutions should be designed to be highly scalable and adaptable to the changing needs of the market.

To help participants get started, they will have access to Alpaca's comprehensive documentation, trading APIs, and development tools. They may also have access to various datasets, market data feeds, and other resources to help them build their solutions.

GOALS AND OBJECTIVES

The overall goal of this project is to create new tools, applications, and integrations around the Alpaca platform and provide users with new ways to trade.
Suggested goal - Use Alpaca's paper-trading API and create trading algorithm on top of it.

To achieve this the following activities are suggested:

- Project Specification
  - Obtain insights into the Alpaca platform.
  - Describe the business case of a proposed concept.
- System Architecture
  - Describe an overview of how a system can be built.
- Development of Minimum Viable Concept
  - Using Figma or Streamlit to demonstrate how the system can be deployed.
- Tips on what possible triggers to include in the trading algorithm
  - Volume of traded stocks
  - Real-time stock price
  - Historical data
  - Buy- and sell triggers

CASE 3 - CLIMATE RISK ASSESSMENT

BACKGROUND

When we talk about "the sustainability of stock trades" we're not typically referring to the act of trading stocks in and of itself. Instead, we're usually discussing the sustainability implications of the companies

behind those stocks. In essence, it's about evaluating the environmental, social, and governance (ESG) factors of the companies whose stocks are being traded. The calculation of ESG (Environmental, Social, and Governance) ratings is complex and can vary significantly between different rating agencies. Each agency usually develops its methodology, which might involve proprietary algorithms, weightings, and metrics. Because of this variation, a company might receive different ESG ratings from different agencies, even though the underlying data might be largely similar.
Here's a general overview of how ESG ratings might be calculated:

Data Collection:

- **Public Disclosures:** Review of annual reports, sustainability reports, and other public documents.
- **Surveys and Questionnaires**: Direct engagement with companies to gather more specific or additional information.
- **Third-party Data:** Information from non-profit organizations, news sources, and other external data providers.
- **On-site Audits:** Some agencies might conduct audits, especially for verifying specific environmental or social practices.
- **Metric Definition and Weighting:** Each agency will define its set of metrics for Environmental, Social, and Governance pillars. Different metrics might have different weightings based on their perceived importance.

Scoring:

Companies are scored on individual metrics based on the data collected. These individual scores are then aggregated, often using weighted averages, to create scores for the Environmental, Social, and Governance pillars. A composite ESG score might then be computed based on the three pillar scores.

Normalization and Peer Comparison:

Scores might be normalized to ensure comparability across companies and industries. Companies are often benchmarked against peers in their industry to give a relative performance rating.

Hypothetical Formula Examples:

1. **Carbon Emissions Score:**

Carbon Score = (Company's Annual Carbon Emmissions / Industry Average Annual Carbon Emissions) * 100
A score of 100 would mean the company's emissions are at the industry average. Below 100 indicates better performance (less emissions), and above 100 indicates worse performance.

1. **Board Diversity Score:**

Diversity Score = (Number of Diverse Board Members / Total Number of Board Members) * 100
This would give a percentage representation of diverse board members.

1. **Employee Satisfaction Score:**

Based on annual employee surveys. If 85% of employees report being satisfied or very satisfied, the Employee Satisfaction Score could be 85.
Hypothetical Composite ESG Score:

Let's say we only consider three factors: Carbon Emissions, Board Diversity, and Employee Satisfaction. We decided to give 50% weight to Carbon Emissions, 25% to Board Diversity, and 25% to Employee Satisfaction.

ESG Score = 0.5 * Carbon Score + 0.25 * Diversity Score + 0.25 * Employee Safisfaction Score

Hypothetical Rating Ranges:

Once scores are normalized and compared against peers, rating agencies could categorize them into different ranges or tiers. A simple tier system could be:

- 80-100: Excellent
- 60-79: Good
- 40-59: Fair
- 20-39: Poor
- 0-19: Very Poor

PROBLEM DESCRIPTION

Your group is asked to create a transparent and reliable ESG scoring system that evaluates companies based on their environmental, social, and governance practices. Participants are tasked with developing their own ESG score, complete with a detailed methodology and sample calculations derived from real-world data.

GOAL

Develop your own ESG score, detail your methodology and offer sample calculations derived from real-world data you've unearthed.

SUGGESTED APPROACH

- Technology exploration
  - Explore available data sources.
- Project Specification
  - Define your own KPIs assessing various parameters of a company's health.
- Method - System Architecture
  - Describe how data can be obtained, processed and aggregated, to keep your ESG score up-to-date
- Development of Minimum Viable Concept
  - Create Figma/Plotly design showcasing all KPIs and the final ESG score

CASE 4 - DATA-DRIVEN DECISIONS IN MERGERS & ACQUISITIONS

BACKGROUND

The merger and acquisition (M&A) landscape is booming. In 2022, global M&A transactions reached an impressive value of $3.4 trillion. Yet, a staggering statistic remains: despite the growth and volume of these transactions, between 70% and 90% fail to add the anticipated value for shareholders. With the rapid evolution of technology, many companies are turning to M&As to acquire innovative assets, intellectual property, and renowned brands to better position themselves in the market. These acquisitions can present uncertainties, especially in the tech-dominated era. To navigate this, many are turning to data analytics, with the latest AI and analytics solutions promising informed decision-making and foresight.

With the wealth of data available, there's a significant opportunity to design a robust scoring system to rate the potential success of an M&A. Much of the current data used for evaluation comes from third-party sources, surveys, or interviews. This data often measures innovation investments (like R&D expenditure) and their resulting outputs (e.g., patents). Our goal is to harness patent data and other relevant datasets to develop a scoring method for M&As. By doing so, we aim to provide stakeholders with a hypothetical score, shedding light on the potential success of an acquisition or merger.

PROBLEM DESCRIPTION

Your team is tasked to design a robust scoring system that can evaluate the potential success of an M&A transaction. Although there are many evaluation methods, a significant part of the data used comes from third-party sources, surveys, or interviews. The team should use patent data and other relevant data sets to develop a scoring methodology for M&A. This can provide stakeholders with a hypothetical score that provides insight into the likelihood of a successful acquisition or merger.

Data is available, but there are still opportunities on how to develop the tools for assessing available data. Datasets aim to capture innovation inputs (e.g. R&D expenditure) and to create practical indicators for measuring innovation outcomes (e.g. through patents). This case is based on using patent data in combination with other sources to develop a data-driven method for assessing mergers and acquisitions.

To develop this method the following approach is suggested.

- Project Specification
  - Identify - Indicators on firm-level innovation activities
    - reference work for finding datasets or metrics.
    - available tools
    - reference projects
- System Architecture
  - Propose a conceptual structure of a possible web scraping solution.
- Development of Minimum Viable Concept
  - Using Figma or Streamlit demonstrate how the system can be deployed.

CASE 5 - SENTIMENT ANALYSIS

Background

In the constantly evolving investment landscape, having an edge can make the difference between a successful or unsuccessful investment strategy. In the era of information abundance, the challenge often lies in sifting through data to extract valuable insights. The development of a sentiment analysis (SA) tool addresses this challenge, enabling investment firms to make more informed decisions based on vast amounts of unstructured data found in news articles, financial reports, and social media platforms. Here's how sentiment analysis is commonly applied in fintech:

- Algorithmic Trading: Some trading algorithms incorporate sentiment analysis to predict stock price movements. By assessing the sentiment of news articles, financial reports, and even tweets related to a particular stock, these algorithms can decide whether to buy, sell, or hold a particular asset.
- Risk Management: Analyzing sentiments from news and social media can help in predicting potential risks associated with an investment. For instance, if there's increasing negative sentiment around a company due to some rumored bad news, it could be a precursor to a potential stock price drop.
- Customer Insights and Personal Finance: Fintech companies use sentiment analysis to gauge customer sentiment towards their products and services. This can help them in tailoring their offerings, improving customer experience, or even predicting market demands.
- Regulatory Compliance and Monitoring: Monitoring public sentiment can also be used to detect anomalies or potential cases of market manipulation, which can be crucial for regulatory bodies or institutions to ensure compliance.

You are on a development team that helps a global investment fund in utilizing Artificial Intelligence (think Chat-GPT) to make better decisions. The investment fund aims to analyze the sentiment of news articles, blog posts, social media posts, financial reports, and other sources of textual information that could influence market dynamics.

PROBLEM DESCRIPTION

Your team is tasked to develop a sentiment analysis tool using advanced NLP (Natural Language Processing) techniques. The tool is to be designed to understand the context, recognize the subtleties of language, identify sarcasm, and differentiate between fact and opinion. This is in order to provide a new perspective on investment decision-making, supplementing traditional financial analysis methods. It allowed the client to make more holistic and informed decisions, contributing to improved portfolio performance.

GOALS

- How “ChatGPT” can be tuned to perform sentiment analysis on alternative data sources eg. financial news articles, social media posts, and other unstructured data sources.
- Can SA be used to identify positive or negative sentiment about a particular stock or market trend.

SUGGESTED APPROACH

- Development of a method for sentiment analysis
- Deployment of the method as a tool – optional
- Eg. Use the openai API and streamlit to test of the method.
- Evaluation of the tool/method
- Evaluate the added business value of this tool
- Effects of a streamlined and automated sentiment analysis,

  - Does it free up valuable time for their analyst?
- Does it identify potential investment risks and opportunities that were not immediately apparent from conventional analysis methods?
- Does it enable earlier detection of shifts in public sentiment that often precede market changes, giving the firm a competitive edge?
- Does it capture and interpret a vast amount of data quickly? (which is impossible manually).
- Recommendations and further work
  - How can (or should) this tool be incorporate into the fund investment strategies?

Examples of sentiment analysis and other cases where ChatGPT is used

CASE 6 - DATA-DRIVEN BANKRUPTCY PREDICTION

BACKGROUND

In the intricate world of investment banking, being able to assess the financial stability and potential risks associated with companies is of paramount importance. While traditional methods provide a myriad of metrics to analyze, synthesizing them into a single, digestible, and actionable score can significantly streamline the decision-making process for stakeholders. This concept is not foreign, as seen with ESG scores focusing on environmental, social, and governance factors. Drawing inspiration from such consolidated metrics, there's a growing demand for a similar, unified "Bankruptcy Score" to gauge a company's risk of insolvency.

PROBLEM DESCRIPTION

Your group is asked to conceptualize, design, and validate a comprehensive "Bankruptcy Score" – a single metric that accumulates a blend of financial, operational, and market-driven indicators to predict a company's risk of bankruptcy. This score should not only simplify the risk assessment process but also enhance its predictive accuracy.

GOALS

Develop your own Bankruptcy score method that presents key aspects such as

- Consolidation of Metrics: Identify, evaluate, and consolidate a range of traditional and alternative data sources into a singular score.
- Dynamic Adaptability: Ensure the score is adjustable based on industry, geography, and size of the company, allowing for more tailored risk assessments.
- Transparency & Interpretability: Make the methodology behind the score transparent and easily interpretable for stakeholders.
- Validation & Reliability: Backtest the score with historical bankruptcy cases to validate its predictive power and reliability.

SUGGESTED APPROACH

In order to achieve this the following activities is suggested

- Technology exploration
  - Explore available data sources.
    - Review of the “US Company Bankruptcy Prediction Dataset” and references projects on how machine learning can be used to predict bankruptcy.
- Project Specification
  - Define your own KPIs assessing various parameters of a company's health.
- Method - System Architecture
  - Describe how data can be obtained, processed and aggregated, to keep your score up-to-date
- Development of Minimum Viable Concept

  - How can this data be presented? (ex. Figma description)
- Suggested limitations
  - The goal of this project could be proposing a method for predicting bankruptcy in a specific sector such as the financial sector such as larger financial institutions to regional banks in the US based on a knowledge transfer from this public available dataset.